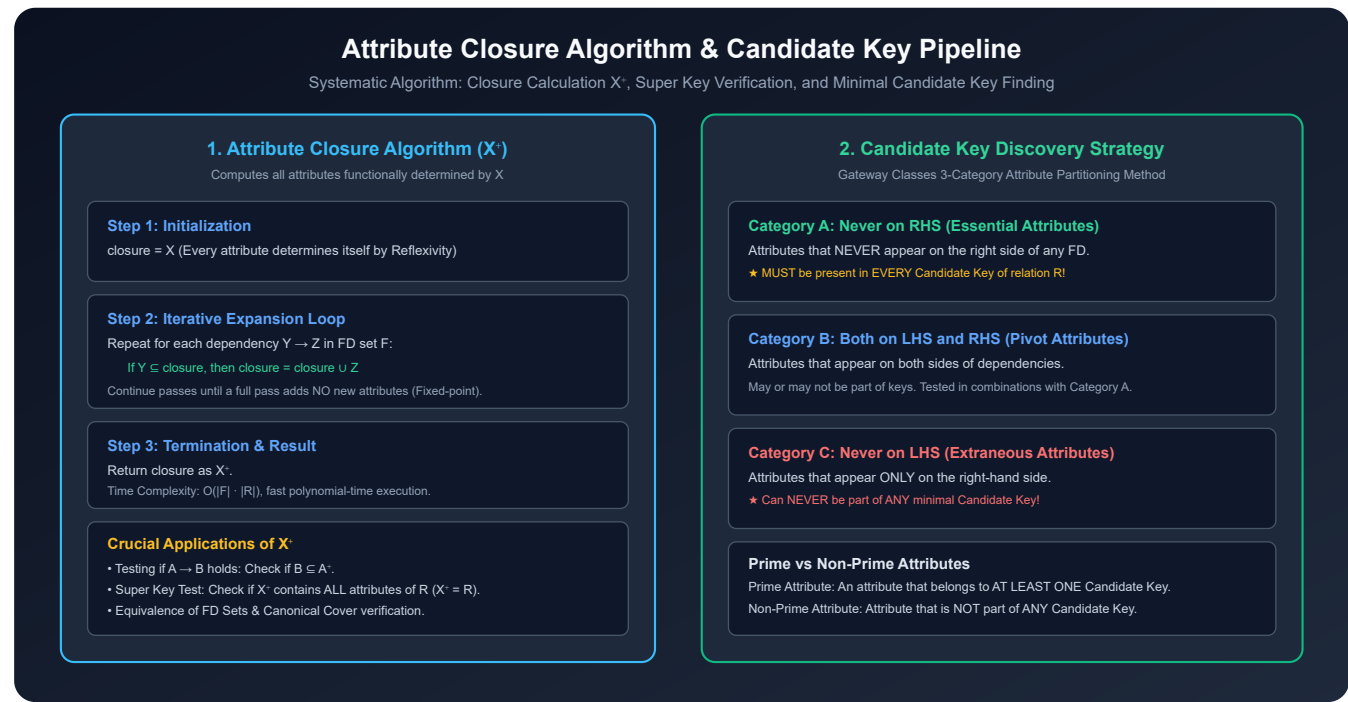


Module 03: Attribute Closure & Candidate Key Algorithms



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Attribute Closure Algorithm (X^+), Applications of Closure, Candidate Key Finding Algorithm, Attribute Categorization (LHS/RHS matrix), Prime vs Non-Prime Attributes, Solved Gateway Numericals.

1. What is Attribute Closure (X^+)?

Relation schema R aur FD set F ke context me, attribute set X ka **Attribute Closure** (X^+) un sabhi attributes ka collection hota hai jo X se functionally determine kiye ja sakte hain using Armstrong's Axioms.

Formal Definition:

$$X^+ = \{ A \mid A \in R \mid \exists F \}$$

Dash X

$$\rightarrow A$$

1.1 The Classical Attribute Closure Algorithm

,

Algorithm: Compute_Attribute_Closure(R, F, X)

Input: Relation schema R , Set of Functional Dependencies F , Attribute set X

Output: Attribute closure X^+

1. Set closure = X ;
2. Repeat:

old_closure = closure;

For each functional dependency ($Y \rightarrow Z$) in F :

If $Y \subseteq \text{closure}$:

closure = closure $\cup Z$;

Until (closure == old_closure);

3. Return closure;

,

Time Complexity: $O(|F| \times |R|)$ — Polynomial time, extremely fast!

2. Core Applications of Attribute Closure in DBMS

1. Testing If a Functional Dependency Holds:

- Problem: Check if $X \twoheadrightarrow Y$
- Method: Compute X^+ . If $Y \subseteq X^+$, then **YES**, the dependency holds; otherwise NO.

2. Super Key Identification:

- If $X^+ = R$ (contains every attribute of the relation), then X is a **Super Key** of R .

3. Candidate Key Identification:

- If X is a Super Key AND no proper subset X' is a super key (minimality condition), then X is a **Candidate Key**!

3. Systematic Candidate Key Finding Algorithm (AKTU Gateway Method)

Jab ek relation me multiple attributes aur 5-6 FDs hoti hain, toh brute force har subset ka closure nikalna exponential (2^n) ho jaata hai. Gateway Classes aur standard database theory ka **3-Category Partitioning Method** candidate keys ko 30 seconds me solve kar deta hai:

Step 1: Divide all attributes of relation R into 3 categories:

Category A (Essential)	Category B (Pivot / Middle)	Category C (Extraneous)
Attributes that NEVER appear on the RHS of dependencies.	Attributes that appear on BOTH LHS & RHS	Attributes that appear ONLY on RHS (Never on LHS).
MUST be in EVERY CK!	May be combined with Cat A.	CANNOT be in ANY CK!

,

3.1 Step-by-Step Procedure

1. Find all attributes belonging to **Category A** (Never on RHS).
 2. Calculate closure of $(\text{Category A})^+$.
- **Case 1:** If $(\text{Category A})^+ = R$, then Category A is the **Unique Candidate Key**! You are finished!
 - **Case 2:** If $(\text{Category A})^+ \neq R$, then combine Category A with single attributes from **Category B**, then pairs, and check which combinations produce a closure equal to R .
3. Never include attributes from **Category C** in candidate key tests.

4. Prime vs Non-Prime Attributes

Yeh distinction 2NF aur 3NF testing ke liye mandatory hai:

- Prime Attribute:** Agar koi attribute relation ke **kisi bhi ek Candidate Key** ka hissa hai, toh use Prime Attribute kehte hain.
- Non-Prime Attribute:** Wo attributes jo relation ke **kisi bhi Candidate Key** me shaamil nahi hote.

Example: If Candidate Keys are $\{AB, BC\}$, then:

- Prime Attributes = $\{A, B, C\}$
- Non-Prime Attributes = All remaining attributes in R .

5. Architectural Diagram

6. Solved Gateway Classes AKTU Numerical Problems

Numerical 1 (From Gateway Slide Deck)

Question: Given relation $R(A, B, C, D, E, F)$ with Functional Dependencies:

$F \rightarrow A$

$B \rightarrow B$

$C \rightarrow C$

$D \rightarrow D$

$E \rightarrow E$

$F \rightarrow F$

Find all Candidate Keys of R , and classify Prime and Non-Prime attributes.

Solution:

1. Analyze Attributes:

- RHS contains: $\{B, C, D, E, F\}$.

- Attribute A never appears on the RHS $\implies$ Category A (Essential) = $\{A\}$.

2. Compute Closure of $\{A\}$:

- $A^+ = \{A\}$

- Since A

$B \in F \implies A^+ = \{A, B\}$

- Since B

$C \in F \implies A^+ = \{A, B, C\}$

- Since C

$D \in F \implies A^+ = \{A, B, C, D\}$

- Since D

$E \in F \implies A^+ = \{A, B, C, D, E\}$

- Since E

$F \in F \implies A^+ = \{A, B, C, D, E, F\} = R$.

3. Conclusion:

- Since $A^+ = R$ and $\{A\}$ has no proper subset, **A is the only Candidate Key.**

- **Prime Attributes:** $\{A\}$

- **Non-Prime Attributes:** $\{B, C, D, E, F\}$

Numerical 2 (Multiple Candidate Keys)

Question: Given relation $R(A, B, C, D)$ with $F = \{AB$

$\rightarrow CD, C$

$\rightarrow A, D$

$\rightarrow B\}$. Find all Candidate Keys.

Solution:

1. Categorization:

- RHS has: $\{A, B, C, D\}$. Every attribute appears on RHS!

- Essential Category $A = \{\emptyset\}$.
- All attributes $\{A, B, C, D\}$ are in Category B.

2. Check pairs:

- $(AB)^+ = \{A, B, C, D\} = R \implies \mathbf{AB}$ is a Candidate Key.

- $(BC)^+$: Since C

$\rightarrow A, (BC)^+ \supseteq \{A, B, C\}$

$\rightarrow D \implies (BC)^+ = R \implies \mathbf{BC}$ is a Candidate Key.

- $(CD)^+$: Since C

$\rightarrow A$ and D

$\rightarrow B, (CD)^+ \supseteq \{A, B, C, D\} = R \implies \mathbf{CD}$ is a Candidate Key.

- $(AD)^+$: Since D

$\rightarrow B, (AD)^+ \supseteq \{A, B, D\}$

$\rightarrow C \implies (AD)^+ = R \implies \mathbf{AD}$ is a Candidate Key.

3. Total Candidate Keys: $\{AB, BC, CD, AD\}$.

4. **Prime Attributes:** $\{A, B, C, D\}$ (Every attribute is prime!).

5. **Non-Prime Attributes:** $\{\emptyset\}$ (None).